OLYMPIC GAMES

DATA ANALYTICS DASHBOARD

Power BI · Interactive Dashboard · Portfolio Project

DATASET	TOOL	COVERAGE
Olympic NPC Medal Records	Microsoft Power BI	1960 – 2018 | 26 Games

Uncovering Patterns · Extracting Insights · Driving Decisions

PROJECT INTRODUCTION

Why Analyze Olympic Data?

The Olympic Games represent the world's largest multi-sport event — a unique intersection of athleticism, geopolitics, economics, and culture.

With records spanning 60+ years and 73 nations, the dataset offers a rare opportunity to apply data analytics to uncover performance patterns, equity gaps, and strategic opportunities that are invisible in raw tables.

This project uses Microsoft Power BI to build a 6-page interactive dashboard that transforms historical records into a decision-support tool for sports bodies, national committees, and policy makers.

Medal Performance

Gold, Silver & Bronze analysis across all nations

Global Coverage

73 countries across every inhabited continent

35 Sport Disciplines

From Athletics to Winter sports, all analysed

1960 – 2018

Six decades of Olympic history in one dashboard

PROJECT OBJECTIVES

01	**Understand the Dataset**	Explore the structure, data types, and quality of the Olympic dataset before building any visualisation.
02	**Identify Patterns & Problems**	Detect medal concentration, geographic disparities in hosting, and imbalances in sport participation.
03	**Extract Meaningful Insights**	Translate raw statistics into clear narratives about country dominance, sport trends, and hosting equity.
04	**Deliver Strategic Recommendations**	Propose data-backed actions for sports governing bodies to address the identified gaps.
05	**Showcase Power BI Capability**	Demonstrate interactive dashboard design including slicers, drill-through, DAX measures, and map visuals.

DATASET OVERVIEW

73	26	16K	35
Countries (NPCs)	Games Editions	Medal Records	Sports

About the Dataset

The dataset records Olympic Games performance from 1960 to 2018, covering both Summer and Winter editions.

Each row represents a unique combination of Games edition, National Olympic Committee (NPC), and sport discipline — with medal counts (gold, silver, bronze) and ranking information.

This structure supports multi-dimensional analysis across time, geography, sport, and nation.

Key Column Groups

Games Dimensions
`games_year, games_city, games_continent, games_season, games_start/end`

NPC (Country)
`npc, npc_name, npc_new, npc_rank, rank_type`

Medal Metrics
`npc_gold, npc_silver, npc_bronze`

Sport
`sport_code, sport`

DATA PREPARATION & MODELLING

1

Data Import & Type Validation

Loaded the CSV dataset into Power BI Power Query. Validated column data types — confirmed medal counts as whole numbers, dates as date fields, and NPC codes as text. Removed erroneous type coercions.

2

Null & Missing Value Handling

Assessed null counts across all 17 columns. Applied conditional replacements for missing NPC names and excluded rows with null medal counts from ranking calculations to prevent skewed KPIs.

3

Calculated Columns & Standardisation

Created a Total Medals calculated column (gold + silver + bronze). Standardised season labels and NPC code formats for consistent cross-page filtering and slicer behaviour.

4

DAX Measures for KPIs

Built DAX measures including Total Countries, Total Medals, Avg Medals per Country, and Medal Share %. These power all KPI cards, trend charts, and the Top 10 Nations ranking bar chart.

TOOLS & TECHNOLOGY

Microsoft Power BI

PRIMARY ANALYTICS & VISUALISATION TOOL

- Power Query — data import, transformation, and type validation
- DAX — calculated columns, KPI measures, and dynamic aggregations
- Interactive slicers: filter by Year, Country, Season across all pages
- Custom visuals: world map, stacked bar, line trend, scatter plot, table
- Cross-page navigation buttons for seamless stakeholder user experience
- 6-page dashboard architecture with consistent theme and branding

Microsoft Excel

Initial data profiling and column-level quality checks before Power BI import

Power Query (ETL)

Data cleaning, null handling, type fixing, and calculated column pipeline

DAX Language

Measures for total medals, share %, avg per country, and running totals

DASHBOARD ARCHITECTURE — 6 Pages

P1

Intro Page

Dataset overview, objectives, navigation guide

P2

Executive Overview

KPI cards, medal trend, top nations, continent share

P3

Country Performance

Medal table, breakdown by country, rank vs medals scatter

P4

Sports Performance

Top / bottom 10 sports, medal distribution by sport

P5

Location Analysis

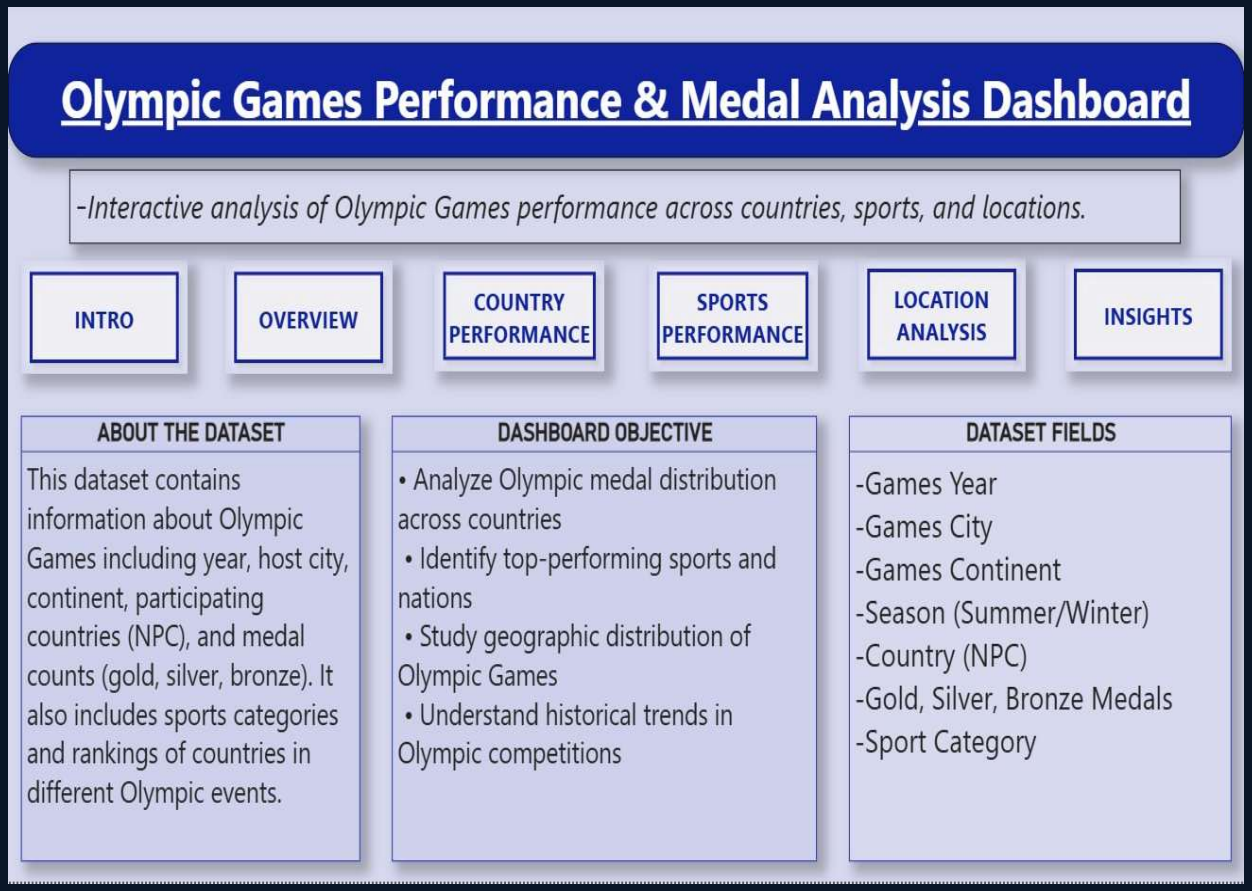
Host city map, games timeline, cities frequency bar

P6

Key Insights & Conclusion

Summary findings, top 5 countries chart, conclusions

DASHBOARD PAGE 1 — Intro & Navigation



Navigation Hub

Six clearly labelled buttons (INTRO → INSIGHTS) let users jump instantly to any dashboard page — critical for non-technical stakeholders.

Dataset Summary

Three information panels cover: About the Dataset, Dashboard Objectives, and Dataset Fields — providing full context before any analysis.

Scope & Objectives

Objectives are stated explicitly: analyse medal distribution, identify top sports, study geography, and understand historical trends.

Field Directory

All 7 key field categories are listed so any user understands the data dimensions being visualised before interacting with filters.

DASHBOARD PAGE 2 — Executive Overview

5 KPI Cards

73 Countries · 26 Games · 16K Medals · 35 Sports · 198 Avg Medals per Country — the entire project scope at a glance.

Medal Trend (1960–2020)

Area-line chart showing medal volume oscillates between Summer and Winter Games. Clear growth from 282 (1960) to peaks above 1,800 in 1980s.

Top Performing Nations

Horizontal bar: USA leads at 2,062 medals, followed by Germany (1,462) and United Kingdom (1,407). Strong western dominance visible.

Medal Share by Continent

Europe dominates with ~6K medals, more than Americas and Asia combined — reinforcing the geographic concentration insight.

KPI ANALYSIS — What the Numbers Reveal

73

Countries

73 NPC nations participated across the 1960–2018 period. Yet the medal table is dominated by fewer than 10 — a 14% participation-to-dominance ratio.

26

Games Editions

26 editions across Summer and Winter Games. Summer Games generate 4–5× more medals per edition than Winter Games.

16K

Total Medals

16,000+ medal records. The top 5 countries account for over 40% of all medals — a stark concentration metric.

35

Sports

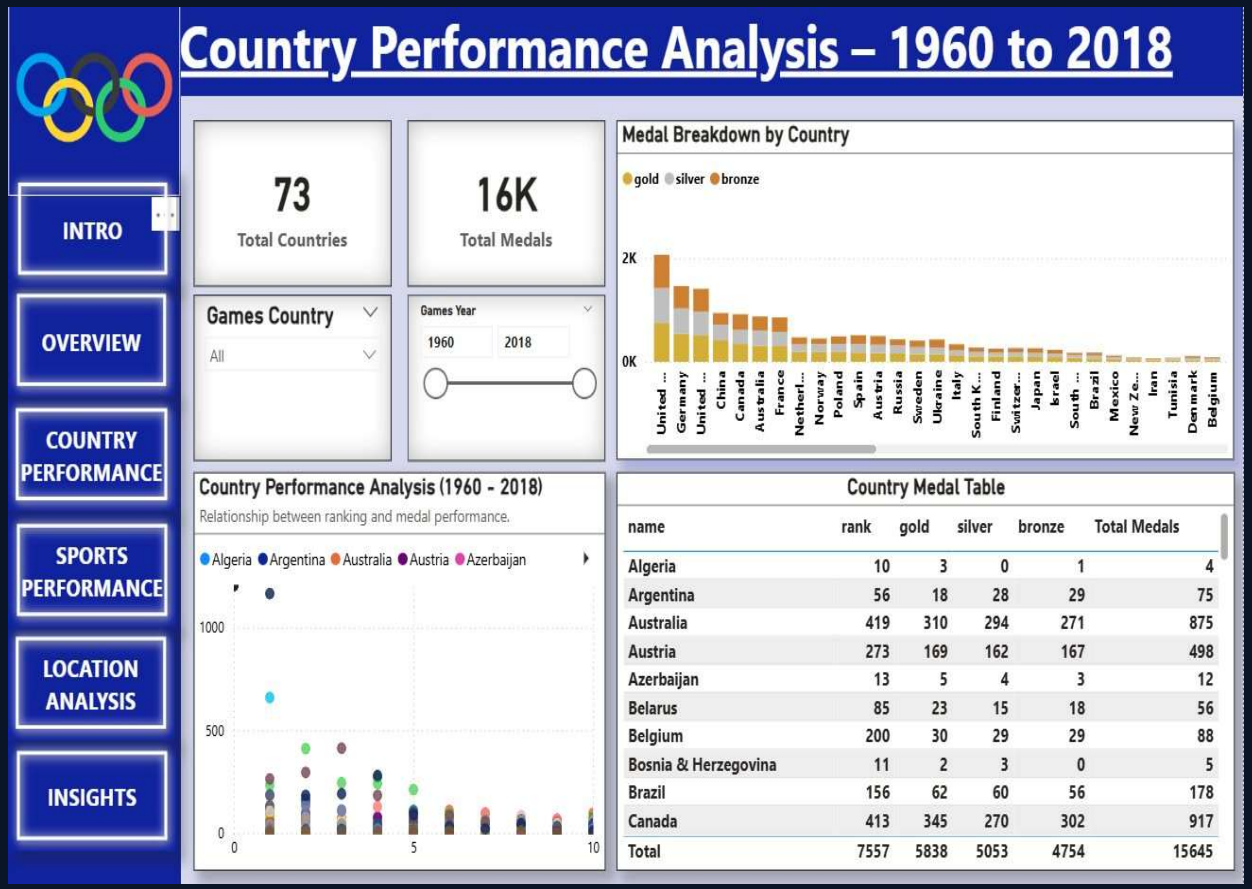
35 sports disciplines tracked. Athletics and Swimming alone contribute the highest medal counts, creating inherent advantages for specialist nations.

198

Avg / Country

Average of 198 medals per country across the full dataset — but the median is far lower, indicating extreme right-skew driven by top performers.

DASHBOARD PAGE 3 — Country Performance Analysis



name	rank	gold	silver	bronze	Total Medals
Algeria	10	3	0	1	4
Argentina	56	18	28	29	75
Australia	419	310	294	271	875
Austria	273	169	162	167	498
Azerbaijan	13	5	4	3	12
Belarus	85	23	15	18	56
Belgium	200	30	29	29	88
Bosnia & Herzegovina	11	2	3	0	5
Brazil	156	62	60	56	178
Canada	413	345	270	302	917
Total	7557	5838	5053	4754	15645

Interactive Filters

Year slider (1960–2018) and Country dropdown allow drill-down into any nation's performance trajectory over time.

Medal Breakdown Chart

Stacked bar by country shows gold/silver/bronze split. USA leads at ~2K total. Visible drop-off after top 3 nations.

Country Medal Table

Full sortable table: Algeria (4 medals, rank 10) to Canada (917 medals, rank 413). Reveals wide rank-to-medals correlation.

Scatter: Rank vs Medals

Country Performance scatter shows clustering of high-rank, low-medal nations — confirming the long tail of underperforming participants.

⚠ ANALYTICAL INSIGHT

Olympic Medals Are Dominated by a Small Number of Nations

TOP 5

COUNTRIES

USA · Germany · UK
China · Canada

account for

>40%

of all 16K medals

Structural Entrenchment

The USA has led the medal table in virtually every Summer Games since 1960. Germany (incl. East/West era) and the UK maintain permanent top-5 positions. This is not random variation — it reflects structural investment in elite sports systems.

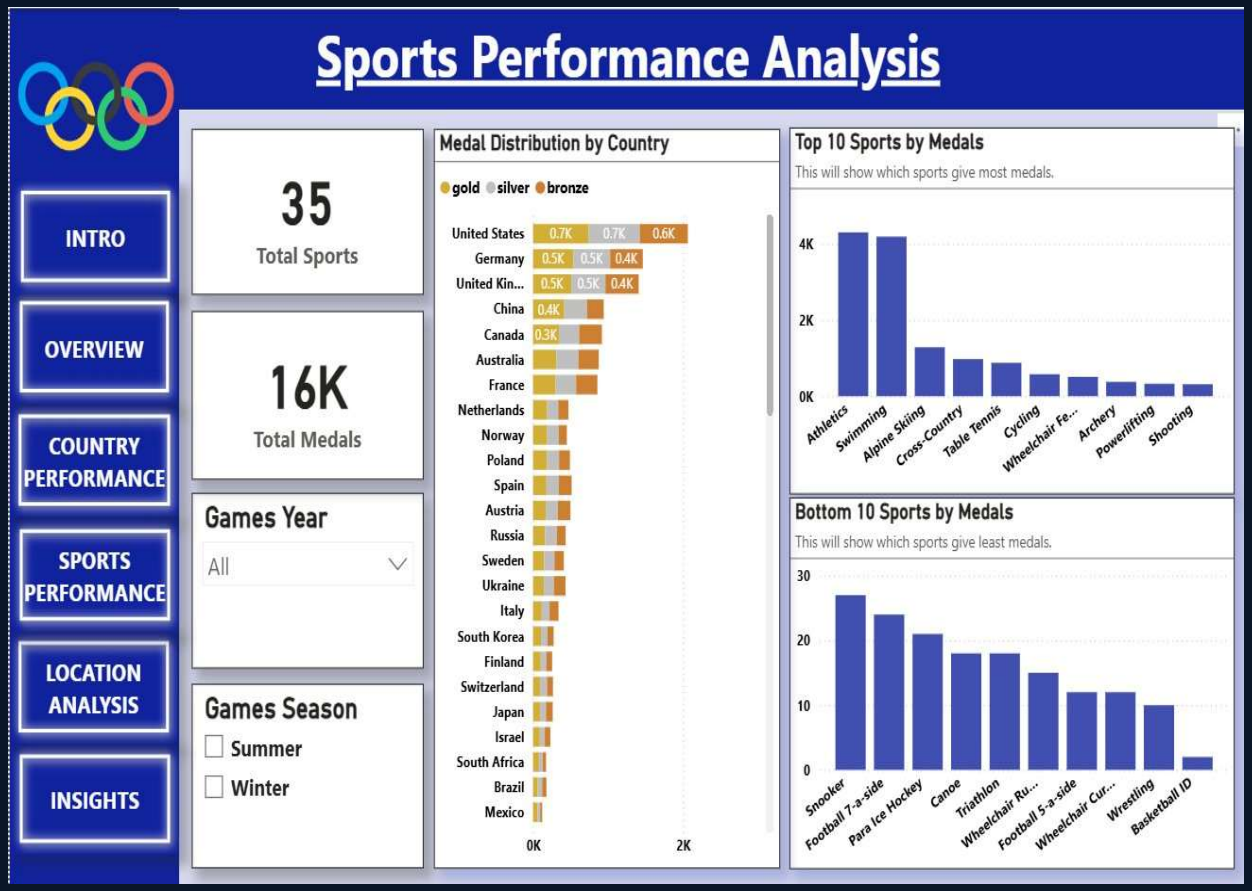
The Long Tail Problem

The Country Medal Table shows dozens of nations with 0–10 total medals across their entire Olympic history. Nations like Algeria (4 medals), Bosnia & Herzegovina (5), and Azerbaijan (12) show minimal returns despite decades of participation.

Compounding Advantage

Nations that win medals attract more sponsorship and government funding — creating a self-reinforcing cycle. Top nations can fund sports academies, sports science, and anti-doping programmes that perpetuate their advantage.

DASHBOARD PAGE 4 — Sports Performance Analysis



35 Total Sports Tracked

Across all Games and disciplines. Summer sports (Athletics, Swimming, Cycling) generate substantially more medals than Winter disciplines.

Top 10 Sports by Medals

Athletics (~4K) and Swimming (~4K) dwarf all other disciplines. Alpine Skiing and Cross-Country follow at ~1K — clear medal concentration in two sports.

Bottom 10 Sports

Snooker, Football 7-a-side, Para Ice Hockey, Canoe and Triathlon each produce <30 medals — revealing severely underrepresented disciplines.

Medal Distribution by Country

USA and Germany lead in medal distribution per country. The stacked bars confirm gold/silver/bronze proportions are consistent among top performers.

⚠ ANALYTICAL INSIGHT

Medal Distribution is Heavily Skewed Toward a Few Dominant Sports

Athletics + Swimming

~8,000 medals combined

from just 2 of 35 sports (~23% of all medals in 2 disciplines)

Bottom 10 Sports

<30 medals each

Snooker, Canoe, Triathlon, Football 7-a-side each award fewer than 30 medals total across all editions

Event Multiplicity Bias

Athletics contains 48+ sub-events (100m, 200m, relay, hurdles…). A nation with elite sprinters can earn 15+ medals. A nation strong in Judo earns 7 at most. The structure inherently rewards generalism.

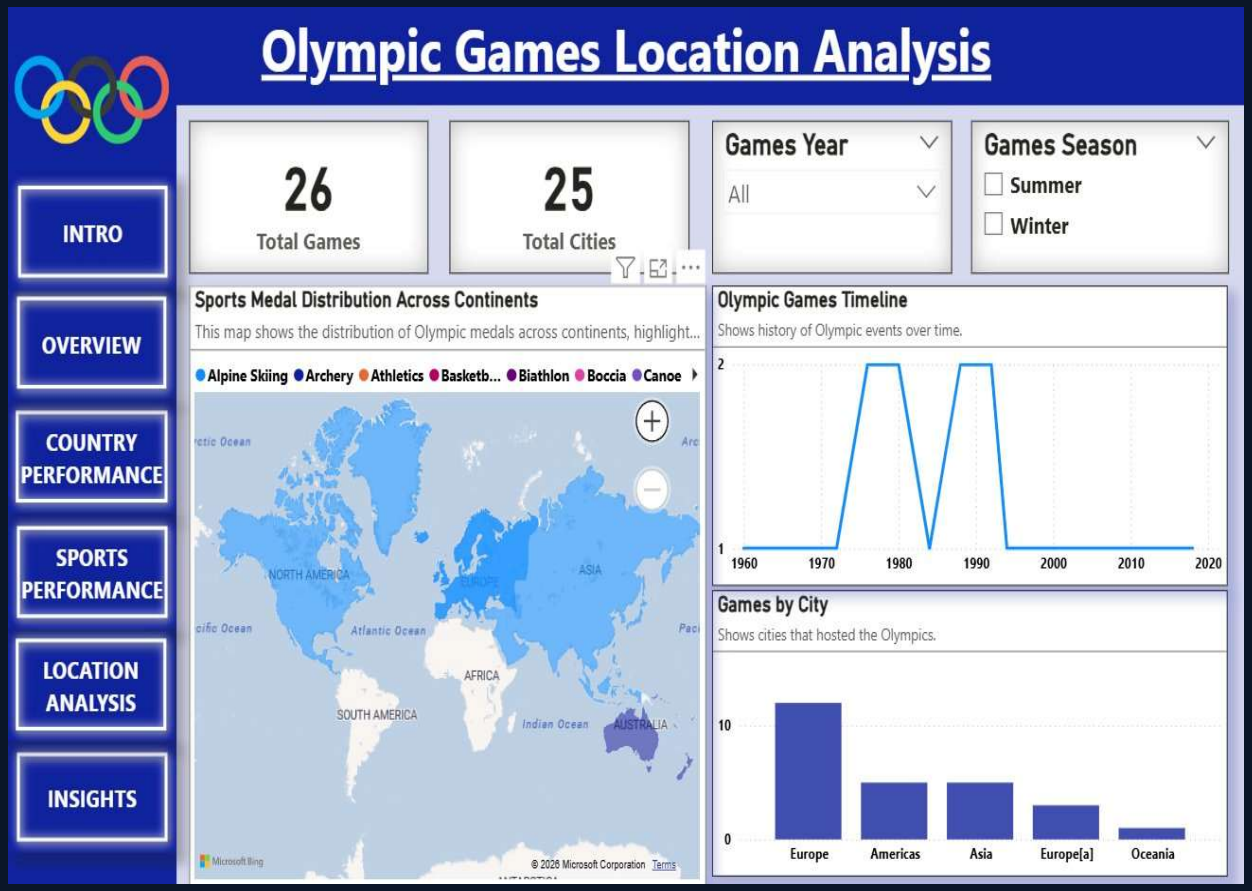
Summer vs Winter Asymmetry

Summer Games generate 4–5× more medals per edition. Nations without winter climates or ice infrastructure are structurally excluded from a large portion of the Games programme.

Sponsor & Broadcast Feedback Loop

High-medal sports attract more TV time and sponsorship, which funds more athletes and infrastructure — widening the gap between dominant and emerging sports disciplines.

DASHBOARD PAGE 5 — Location Analysis



26 Games · 25 Cities

The dataset covers 26 editions across 25 unique host cities, with London being the only city to have hosted more than once in this period.

World Map — Medal Distribution

Chloropleth map shades continents by medal volume. Europe and North America show the darkest shading. Africa and South America are visibly pale.

Games Timeline

Line chart shows alternating Summer/Winter editions. The dual-peak pattern in 1980 and 1992 reflects years with both Summer and Winter Games hosted.

Games by City (Bar)

Europe hosts ~12 Games vs Americas ~4 vs Asia ~4. Oceania and Europe[a] regions each host 1–2 times. Africa: zero.

⚠ ANALYTICAL INSIGHT

Olympic Hosting Rights Are Concentrated in Europe and the Americas

~12	~4	~4	~2	0
Games Hosted	Games Hosted	Games Hosted	Games Hosted	Games Hosted
Europe	**Americas**	**Asia**	**Oceania**	**Africa**

Africa Has Never Hosted

Across the entire 1960–2018 dataset period, no African city has hosted the Olympic Games. With 1.4 billion people and the world's fastest-growing youth population, this is the most significant equity gap in the location data.

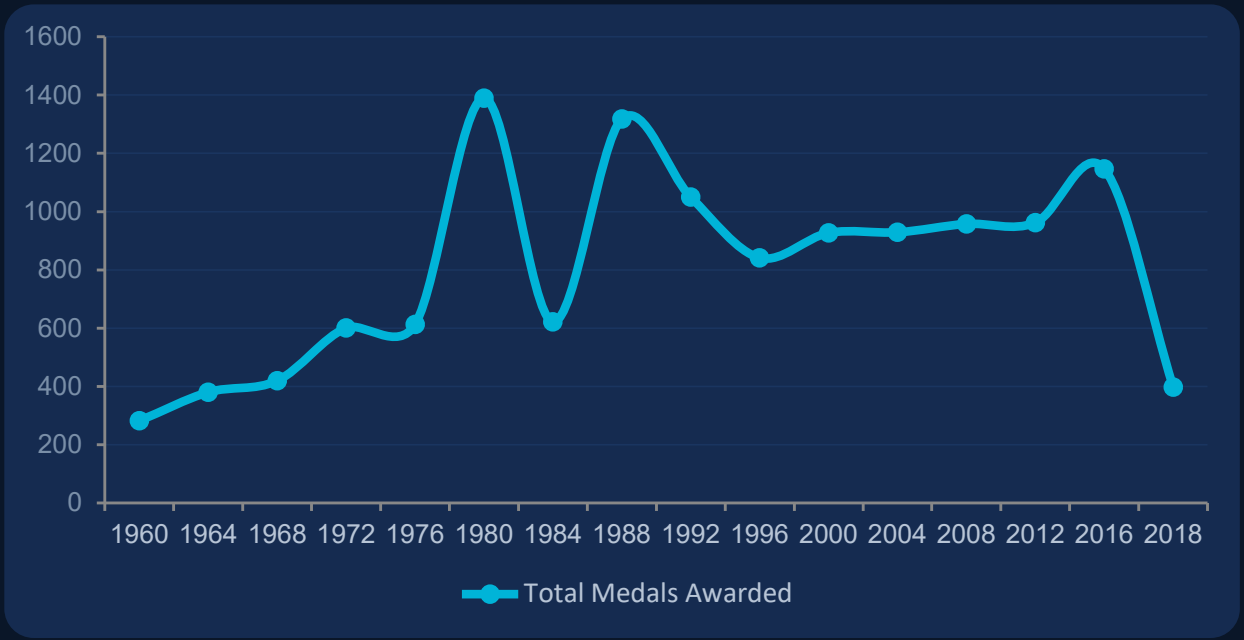
Hosting Drives Performance

Research shows host nations consistently outperform their historical medal average in the year they host. By never hosting, African nations miss this performance catalyst — compounding their medal disadvantage.

Economic Benefits Concentrated

The multi-billion-dollar infrastructure investment and tourism boost from hosting accrues almost entirely to Europe and North America — nations that already benefit most from Olympic success.

TRENDS OVER TIME — 1960 to 2018



Medal Trend: Oscillation Driven by Summer/Winter Alternation

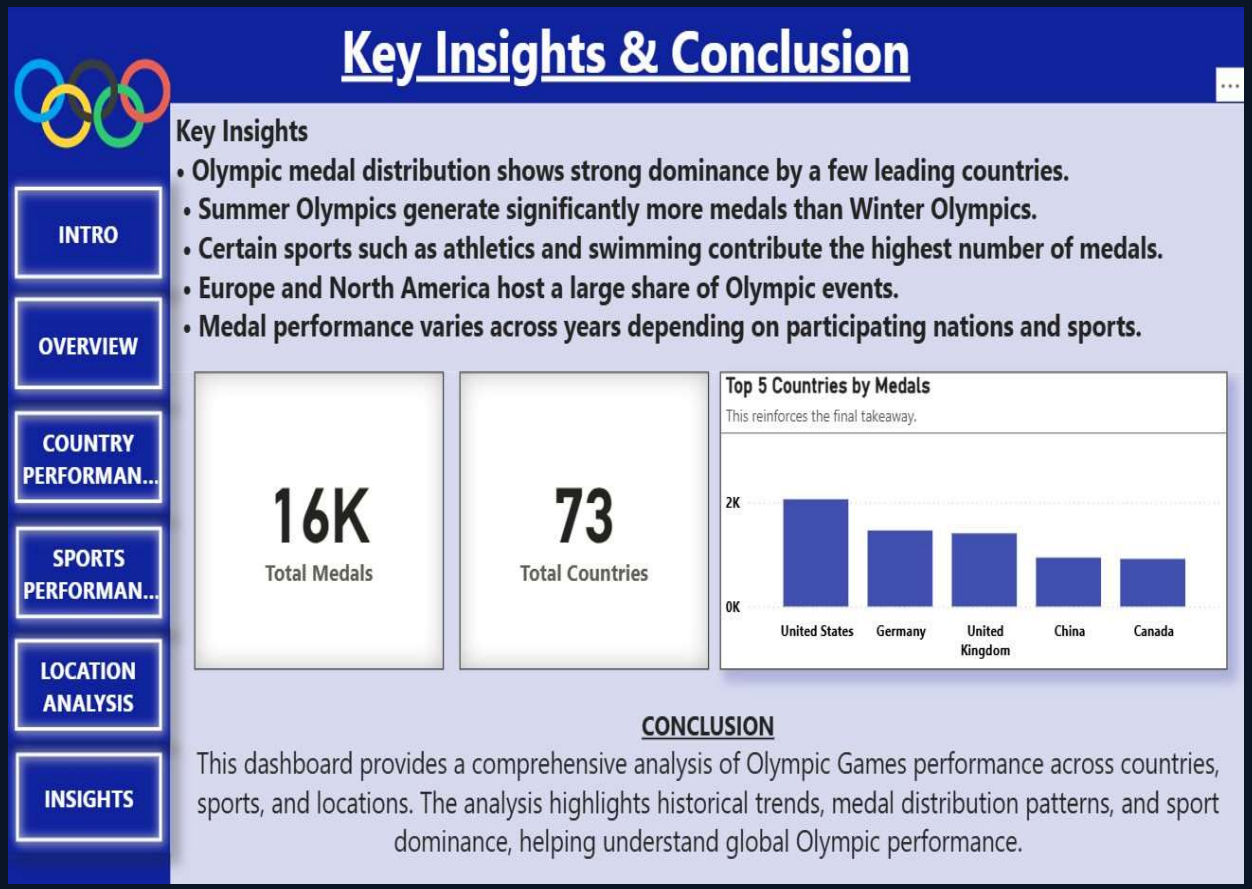
Summer/Winter Oscillation
The medal count alternates dramatically — Summer Games (e.g. 1980: 1,389) produce 4–5× more medals than Winter Games (e.g. 2018: 398). The trend chart visualises this structural pattern clearly.

Long-Term Growth (1960→2016)
Overall medal volume grew from 282 (1960) to a peak of 1,389 (1980) and settled around ~950–1,150 in the modern era, reflecting expanded sport programmes.

Participation Expansion
New nations began competing post-1991 following Soviet dissolution — adding former Soviet republics as independent NPCs, which temporarily increased both participation and medal diversity.

Stability in Top Performers
Despite changing participation, the top 5 nations remain largely constant across all decades — confirming that dominance is structural rather than cyclical.

DASHBOARD PAGE 6 — Key Insights & Conclusion



5 Key Findings

Dashboard confirms: medal dominance by few nations; Summer/Winter gap; Athletics/Swimming lead; Europe/Americas hosting concentration; year-on-year medal variation.

KPI Reinforcement

16K total medals and 73 countries repeated as anchor KPIs to frame all five findings — reminding viewers of the scale of the analysis.

Top 5 Countries Chart

Bar chart confirms USA (2K+) → Germany → UK → China → Canada in final rankings. Consistent with country performance page findings.

Conclusion Statement

Dashboard provides comprehensive analysis of Olympic performance across countries, sports, and locations — explicitly connecting the visualisation to decision-support value.

DATA-DRIVEN RECOMMENDATIONS

1

Invest in High-Medal Sports

National Olympic Committees

Countries seeking to improve overall medal rankings should concentrate resources on Athletics and Swimming — the two disciplines that generate the most medals. A national academy focused on these sports offers the highest return on sports investment.

2

Emerging Nations: Target Niche Sports

Developing Nation Sports Bodies

Nations with limited budgets should identify lower-competition sports (Table Tennis, Shooting, Archery) where fewer competitors and lower infrastructure costs improve medal probability. Data shows several sub-4K-medal sports with high win rates for developing nations.

3

Expand Olympic Hosting Opportunities

IOC & Regional Committees

The IOC should actively develop an African hosting roadmap. Hosting has a proven positive impact on domestic sports participation, infrastructure, and medal performance. Africa's zero-hosting record represents the single largest equity gap in the dataset.

4

Use Data Analytics for Sports Policy

IOC & Sports Governance Bodies

This Power BI dashboard demonstrates that historical Olympic data can drive forward-looking policy decisions. IOC committees should adopt interactive dashboards as standard tools for sport programme reviews, funding allocation, and host city selection.

CONCLUSION & PROJECT IMPACT

This project demonstrates that Olympic Games data, when analysed through a well-structured Power BI dashboard, can surface meaningful patterns, identify equity gaps, and generate strategic recommendations — transforming historical records into a genuine decision-support tool.

3 Problems Identified

- Medal concentration: top 5 nations = 40%+ of medals
- Sport imbalance: 2 sports generate ~23% of all medals
- Hosting exclusion: Africa has never hosted Olympic Games

4 Recommendations

- Invest in high-medal sports (Athletics, Swimming)
- Emerging nations: target niche disciplines strategically
- Expand hosting rights to underrepresented continents
- Adopt Power BI dashboards for sports policy decisions

6 Dashboard Pages

- Intro · Overview · Country · Sports · Location · Insights
- Interactive slicers, maps, trend lines, scatter plots
- DAX measures + Power Query ETL pipeline
- Professional portfolio-ready case study

THANK YOU

Olympic Games Data Analytics · Microsoft Power BI · Portfolio Project

PROJECT SUMMARY

- 6-page interactive Power BI dashboard built on Olympic Games dataset (1960–2018)
- 73 countries · 26 games · 16,000+ medals · 35 sports disciplines analysed
- 3 analytical problems identified with 4 data-driven strategic recommendations
- DAX measures, Power Query ETL, interactive slicers, and world map visualisations

Questions & Discussion Welcome · Data tells the story — let's explore it together